

ANNEX A

ItikCare Prescriptive Recommendations
Rules / Threshold Table

ItikCare Prescriptive Recommendations — Rules/Threshold Table

1st Pass: Atomic Rules

1.1 Flock Age (flock_age_weeks)

Feature	Tier	Condition
Flock average age (flock_age_weeks)	HIGH <i>(retire / replacement age)</i>	≥ 80 weeks
Flock average age (flock_age_weeks)	MEDIUM-HIGH <i>(post-peak decline)</i>	> 52, < 80 weeks
Flock average age (flock_age_weeks)	MEDIUM <i>(peak laying)</i>	> 26, ≤ 52 weeks
Flock average age (flock_age_weeks)	MEDIUM-LOW <i>(onset of laying)</i>	> 18, ≤ 26 weeks
Flock average age (flock_age_weeks)	LOW <i>(pre-lay / immature)</i>	≤ 18 weeks

Note: 80 weeks reflects maximum intensive-system laying span; actual on-farm culling may occur earlier, 72–80 weeks, depending on flock performance data.

1.2 Temperature (temperature_c)

Feature	Tier	Condition
Temperature (temperature_c)	HIGH <i>(severe heat stress)</i>	≥ 33°C
Temperature (temperature_c)	MEDIUM-HIGH <i>(elevated heat stress)</i>	> 27°C, < 33°C
Temperature (temperature_c)	MEDIUM/NORMAL <i>(thermoneutral comfort zone)</i>	> 23°C, ≤ 27°C
Temperature (temperature_c)	MEDIUM-LOW <i>(mild cold stress)</i>	> 18°C, ≤ 23°C
Temperature (temperature_c)	LOW <i>(cold stress risk)</i>	≤ 18°C

1.3 Humidity (humidity_pct)

Feature	Tier	Condition
Humidity (humidity_pct)	HIGH <i>(severe moisture stress)</i>	≥ 88%
Humidity (humidity_pct)	MEDIUM-HIGH <i>(elevated humidity)</i>	> 70%, < 88%
Humidity (humidity_pct)	MEDIUM/NORMAL <i>(in range)</i>	≥ 50%, ≤ 70%
Humidity (humidity_pct)	MEDIUM-LOW <i>(mild dryness)</i>	> 40%, < 50%
Humidity (humidity_pct)	LOW <i>(dry conditions)</i>	≤ 40%

1.4 Feed Intake per Bird ($\text{feed_intake_kg} \div \text{flock_size}$)

Note: Feed intake uses only three outcome tiers per age group. Converted to gram.

Feature	Age Tier	Tier	Condition
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)	LOW	LOW (underfed)	< 70 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)		NORMAL (on track)	≥ 70 , < 120 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)		HIGH (overfed)	≥ 120 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)	MEDIUM-LOW	LOW (underfed)	< 110 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)		NORMAL (on track)	≥ 110 , < 150 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)		HIGH (overfed)	≥ 150 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)	MEDIUM	LOW (underfed)	< 115 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)		NORMAL (on track)	≥ 115 , < 160 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)		HIGH (overfed)	≥ 160 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)	MEDIUM-HIGH	LOW (underfed)	< 110 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)		NORMAL (on track)	≥ 110 , < 150 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)		HIGH (overfed)	≥ 150 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)	HIGH	LOW (underfed)	< 90 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)		NORMAL (on track)	≥ 90 , < 130 g/bird/day
Feed intake per bird ($\text{feed_intake_kg} \div \text{flock_size}$)		HIGH (overfed)	≥ 130 g/bird/day

Note: For the LOW age tier, the NORMAL feed intake tier of 70 g/bird/day floor applies to ducklings from Week 6 onward; brooding-phase ducklings (Weeks 1–5) will consume less than this by design and should not be flagged as underfed.

2nd Pass: Combination and Prescriptive Rules

2.1 Flock Age × Feed Intake (Status & Recommendation)

Rule	Age Tier	Feed Result	Status	Preventive Recommendation
AF1	Low	Low	Underfed	Increase toward 70–110 g/bird/day; underfeeding at this stage delays growth and laying maturity.
AF2	Low	Normal	On Track	Maintain current ration; monitor weekly weight gain.
AF3	Low	High	Overfed	Reduce ration; excess protein risks leg-development issues in growing ducklings.
AF4	Medium-Low	Low	Underfed	Increase toward 110–130 g/bird/day; risk of delayed/weak start to laying.
AF5	Medium-Low	Normal	On Track	Maintain current ration through the transition into laying.
AF6	Medium-Low	High	Overfed	Reduce ration; excess feed at this stage has no added production benefit.
AF7	Medium	Low	Underfed	Increase toward 115–150 g/bird/day promptly; peak-age underfeeding directly reduces egg output/size.
AF8	Medium	Normal	On Track	Maintain current ration through peak production.
AF9	Medium	High	Overfed	Reduce ration; excess feed at peak is wasted, not converted to yield.
AF10	Medium-High	Low	Underfed	Increase toward 110–140 g/bird/day to avoid unnecessary output loss.
AF11	Medium-High	Normal	On Track	Maintain current, appropriately reduced ration.
AF12	Medium-High	High	Overfed	Reduce ration; feed cost return is falling as production declines.
AF13	High	Low	Underfed	Increase toward 90–120 g/bird/day; even spent layers need a maintenance minimum.
AF14	High	Normal	On Track	Maintain current maintenance-level ration.
AF15	High	High	Overfed	Reduce ration; consider evaluating flock replacement given minimal remaining output.

2.2 Temperature Flag (temperature tier × humidity tier)

Note: Rows = temperature tier; columns = humidity tier.

Temp ↓ / Humidity →	High	Medium-High	Medium/Normal	Medium-Low	Low
High (≥33°C)	High	High	High	Medium-High	Medium-High
Medium-High (27–33°C)	High	Medium-High	Medium-High	Medium-High	Medium/Normal
Medium/Normal (23–27°C)	Medium-High	Medium/Normal	Medium/Normal	Medium/Normal	Medium-Low
Medium-Low (18–23°C)	Medium-High	Medium-Low	Medium-Low	Medium-Low	Medium-Low
Low (≤18°C)	Medium-High	Medium-Low	Low	Low	Low

2.3 Humidity Flag (humidity tier × temperature tier)

Note: Rows = humidity tier; columns = temperature tier.

Humidity ↓ / Temp →	High	Med-High	Med/Normal	Med-Low	Low
High (≥88%)	High	High	High	High	High
Medium-High (70–88%)	High	High	Medium-High	Medium-High	Medium-High
Medium/Normal (50–70%)	Medium/Normal	Medium/Normal	Medium/Normal	Medium/Normal	Medium/Normal
Medium-Low (40–50%)	Medium/Normal	Medium/Normal	Medium-Low	Medium-Low	Medium-Low
Low (≤40%)	Medium-Low	Medium-Low	Low	Low	Low

2.4 Recommendation Text — Temperature Flag

Result	Preventive Recommendation
High	Severe heat stress risk. Increase ventilation/fans immediately, ensure shade and cool water/bathing access, consider roof misting. Expect reduced feed intake and a likely drop in egg yield forecast.
Medium-High	Elevated heat stress. Improve airflow, check shade/water access adequate for flock size, monitor feed intake.
Medium/Normal	Within acceptable comfort range. Maintain current housing ventilation and shade setup.
Medium-Low	Mild cold stress risk. Reduce drafts, check windbreaks; ducklings may need supplemental warmth.
Low	Cold stress risk. Provide draft-free, insulated housing and supplemental heat, especially for ducklings.

2.5 Recommendation Text — Humidity Flag

Result	Preventive Recommendation
High	Severe moisture stress. Increase ventilation urgently, clear wet litter, check drainage, consider reducing stocking density.
Medium-High	Elevated humidity. Improve ventilation, inspect litter condition, watch for respiratory signs.
Medium/Normal	Within acceptable range. Maintain current housing ventilation.
Medium-Low	Slightly dry, generally not harmful. Ensure water access, especially if paired with heat.
Low	Very dry. Monitor water intake if paired with heat; lower priority on its own.

2.6 Recommendation Text — Flock Age

Note: Flock Age does not require a second factor to combine with, Pass 1 tier (Section 1.1) maps directly to a recommendation.

Tier	Preventive Recommendation
High	Retire or cull declining layers and shift fully to the replacement cohort. Holding spent birds past this point mostly adds feed cost with little egg return.
Medium-High	Start raising a replacement cohort now — this is the actionable preventive window. New layers typically need 22–26 weeks from hatching to first lay depending on strain and management conditions, so initiate orders and brooding now to avoid a production gap.
Medium/Normal	No replacement action needed; focus on protecting production during this peak/sustained laying window.

Medium-Low	Confirm laying has started on schedule (~22–23 weeks); a noticeable delay may signal a nutrition or health issue.
Low	No retirement/replacement action relevant yet — flock is still growing and not expected to lay.

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